

Group Number:
Student Names:
Marks: 4 points

COMP 251: Data Structures & Algorithms
Group Assignment 1: Java Primer and Object-oriented design



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******* INSTRUCTIONS *******

- Read each task carefully before starting. Make sure you understand the task requirements, expected output format, and any constraints mentioned.
- All programming tasks must be implemented in **Java**. Ensure that your code compiles and runs correctly using a standard Java development environment (recommendation: **BlueJ**).
- Use meaningful variable and method names to improve readability and understanding of your Java code.
- Provide clear, well-structured Java code and a corresponding screenshot of the output for each programming task.
- Where applicable, provide clear written explanations to justify your answers.
- All submitted work must be your own (or your group's) original work. Any form of plagiarism or unauthorized collaboration will be dealt with in accordance with university policies.

Task 1: StringBuilder class

Write a Java class in which an instance of a **StringBuilder** class is created. Then, use `append()` method of this class at-least four times to print the following string "Hello, name! Welcome to COMP 251 course." Note that you can replace "name" with your name in this string.

Task 2: Wrapper classes

Write a Java program in which two integer numbers are represented in the form of Strings, e.g.,

```
1 String s1 = "135";  
2 String s2 = "215";
```

and this program should be able to print the sum of these two numbers. In particular, the output should be like this:

Output:

Sum of 135 and 215 is: 350

Task 3: == vs. a.equals(b)

In the lecture, we learnt about the difference between '==' and 'a.equals(b)'. Consider the following code fragment

```
3
4 String str1 = "Hello";
5 String str2 = str1;
6 String str3 = new String("Hello");
7
8 boolean x = (str1 == str2);
9 boolean y = (str1 == str3);
10 boolean z = str1.equals(str3);
```

What are the values of variables 'x', 'y' and 'z' in lines 8 -10? Provide rationale of your answer.

Task 4: Input using the java.util.Scanner Class

Write a Java program that takes an integer input from the user and determines whether the entered number is even or odd. For example, if a user enters '16', then output should be following:

Output:

```
Enter an integer: 16
16 is an even number.
```

Task 5: Modify the CreditCard Class

In the lecture, we have gone through a CreditCard Class. In this task, modify the CreditCard class to add a public String toString() method that returns a String representation of the card. Also, modify the main method from Code Fragment 1.6 accordingly so that the main method uses the standard println command instead of calling static method in Line 17 of the Code Fragment 1.6.

Note that the output should remain the same as given in the Code Fragment 1.7 of the textbook.

Task 6: Java's Generics Framework for Key-Value Pair Class

In the lecture, we learnt about using the Java's Generics Framework (e.g., refer to slide 81). In this task, you are required to create a *generic* key-value pair class that can store pairs of data with different types for keys and values. The class should have the following methods:

- Pair(K key, V value): Constructor to initialize a pair with a key of type K and a value of type V.
- K getKey(): Return the key.
- V getValue(): Return the value.
- void setKey(K key): Set a new key.
- void setValue(V value): Set a new value.

- `String toString()`: Return a string representation of the pair, e.g., “Key: Value”.

Moreover, implement a main method that demonstrates the use of your generic Pair class. Test it with pairs of different data types for keys and values, such as (String, Integer), (String, Double), and (Integer, String). For instance, one of the possible outputs is following:

Output:

Key: Age, Value: 25
 Key: Price, Value: 19.99
 Key: 1, Value: One
 Key: Updated Age, Value: 30 // after invoking setValue and setKey methods

Task 7: Inheritance and **super** reserved word

Consider implementations of the following three classes, A, B and C

```

12 public class A {
13     int x = 1;
14     public void setIt(int y) { x = y; }
15     public int getIt( ) { return x; }
16 }
17 public class B extends A {
18     int x = 2;
19     public void setIt (int y) { x = y; }
20     public int getIt( ) { return x; }
21     public void superSetIt (int y) { super.x = y; }
22     public int superGetIt( ) { return super.x; }
23 }
24
25 public class C extends B {
26     int x = 3;
27     public void setIt (int y) { x = y; }
28     public int getIt( ) { return x; }
29     public void superSetIt (int y) { super.x = y; }
30     public int superGetIt( ) { return super.x; }
31     public void superDuperSetIt(int y) { super.superSetIt(y); }
32     public int superDuperGetIt( ) { return super.superGetIt( ); }
33     public static void main(String[ ] args) {
34         C c = new C( );
35         c.superDuperSetIt(4);
36         System.out.println("C's is " + c.getIt( ));
37         System.out.println("B's is " + c.superGetIt( ));
38         System.out.println("A's is " + c.superDuperGetIt( ));
39     }
40 }

```

What is the output? Provide a **thorough justification** of your answer.